CASE STUDY

Titan – only moon with clouds and atmosphere.

Titan is Saturn's largest moon and the second largest (after Ganymede of Jupiter) in the solar system. It is the only moon in the solar system with clouds and a dense, planet-like atmosphere.

Scientists believe that conditions on Titan are similar to Earth's early years (the main difference is that, because it is closer to the sun, Earth has always been warmer). According to NASA, "In many respects, Titan, is one of the most Earth-like worlds we have found to date."

Titan was discovered by Dutch astronomer Christiaan Huygens in 1655. The Huygens lander probe sent to the moon aboard NASA's Cassini spacecraft by the European Space Agency is named in his honor. Huygens was the first human-built object to land on Titan's surface. Diameter:5,150 kilometres, about half the size of Earth and almost as large as Mars. Surface temperature: - 179 degree Celsius, which makes water as hard as rocks and allows methane to be found in its liquid form. Surface pressure: Slightly higher than Earth's pressure. Earth's pressure at sea level is 1 millibar while Titan's is 1.6 millibars. Orbital period: 15,945 days. Titan's mass is composed mainly of water in the form of ice and rocky material. Titan has no magnetic field.

2.12 Meteors

There is a bright streak of light flashing seen often in the sky during night for a few seconds. They are called as **'shooting stars'**. They are the removed pieces of rocks mainly from the Asteroid belt. They are called **Meteoroids** before they enter into our atmosphere. They enter into the atmosphere with great speed. But most of them are burnt when they enter into the atmosphere.

After entering into our atmosphere they are called as **Meteors**. Some pieces do not burn fully and they fall on the earth and make craters. The large unburned pieces of rocks that fall on the earth are called **Meteorites**.

Examples for Meteorite Fall: Meteor crater in Northern Arizona and Lake Lonar in Buldhana District of Maharastra in India were created by meteor impacts.

2.13 Shape and size of the Earth

It once was believed that the Earth was flat and that ships could sail over the edge. This view persisted even in the middle ages and was an issue in recruitment of Columbus.

Early Greek view was that the world was surrounded by the ocean (*Oceanus*), origin of all rivers. Anaximander (600 B.C) proposed that cylindrical earth was surrounded by celestial sphere. Pythagoras (582-507 B.C.) believed that the Earth was a sphere, which was considered the most harmonious geometric shape. Aristotle (384-322 B.C.) described observations that supported the theory that the Earth was a sphere. These included the fact that the shadow of the moon is circular in lunar eclipses and constellations were higher in the sky as one travelled south. Eratosthenes

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